

FLCR-08 - Discrete-Continuous Bridge Without Physical Overclaim

Series: Formalizing LCR, Unifying Standard Models **Artifact:** formal paper source **Status:** first-pass rich rewrite; requires final citation and build pass. **Claim posture:** maximal local claim posture with explicit lane boundaries.

Abstract

This paper defines a bridge from finite receipt data to continuous presentation and back to discrete representative form. Piecewise linear interpolation preserves samples; it does not create dynamics. Retraction-style admission into MDHG, SpeedLight, or CAM surfaces must preserve provenance, error budgets, and idempotence.

Keywords

interpolation; retraction; quantization; sample preservation; CAM admission.

External Reader Orientation

An outside reviewer should read this paper as a formal-system construction paper. Its local object is **Discrete-Continuous Bridge Without Physical Overclaim**. The paper's immediate contribution is: Defines presentation bridges, sampling obligations, and continuum guards.

The nearest external anchors are finite-state reasoning, typed interfaces, proof-carrying records, conservative extension, and ordinary mathematical citation discipline. LCR terms are therefore internal labels for the construction unless a row explicitly imports an external theorem, citation, dataset, or calibration target.

It does not ask the reader to accept any Standard Model or literal-physics identification at this layer. Such mappings are deferred to the translation papers and must carry their own evidence.

Reviewer Compression

Core object. Discrete-Continuous Bridge Without Physical Overclaim.

Primary result. This paper contributes the local formal object and its safe downstream-use rule.

Primary non-result. This paper does not claim direct identity with Standard Model or physical objects.

Review strategy. Evaluate the formal claim rows first, then inspect the receipt/citation/calibration rows that support each claim. Appendices provide routing and implementation context; they should not be read as stronger than their lane permits.

Peer Reviewer Assessment

Reviewer assumption: the reviewer has been briefed on the FLCR structure, claim lanes, CAM/crystal routing, forge/action surfaces, and the distinction between internal formal results and external physical calibration.

Recommendation. major revision, technically promising.

Review score vector. orientation=2, claim_granularity=2, evidence_visibility=2, falsifier_visibility=2, journal_apparatus=1, base_layer_boundary=2.

Formal claim rows reviewed. 3.

Reviewer Strength Finding

The manuscript is now readable as a bounded formal-system paper: it identifies its local object, separates internal construction from physical interpretation, and exposes claim lanes for review.

Major Revision Requests

- Convert the strongest proof sketch into numbered definitions, lemmas, and propositions with explicit dependencies.
- Replace placeholder source classes with final citation keys, receipt hashes, or dataset identifiers wherever possible.

- Move repeated pass-derived material into a compact evidence appendix and keep the main body focused on the paper-local argument.
- Add at least one worked example showing the local object, legal operation, receipt, residue, and downstream import rule.

Minor Revision Requests

- Normalize theorem capitalization and punctuation.
- Add final figure/table captions where the text currently describes diagrams schematically.
- Ensure every acronym and local term appears in the paper-local glossary.

Acceptance Blockers

- A reviewer can accept the formal contribution without accepting later physics claims, but only if final citations and receipt identifiers are attached.
- The proof sketch must be promoted into a formal proof or explicitly labeled as a construction argument.

1. Contribution And Scope

- Formalizes sample-preserving piecewise-linear presentation.
- Names the boundary between interpolation and physical dynamics.
- Frames MDHG/SpeedLight-style admission as retraction to addressable discrete representatives.
- Adds error-budget requirements for any stronger numerical or physical bridge.

This paper belongs to the LCR-native construction band. Physics and Standard Model language, when it appears, is only a deferred mapping cue, analogy, or explicitly bounded calibration target. The paper's base object must remain stable enough for later papers to cite without importing a stronger physical claim by implication.

2. Source Routing

Primary routed sources:

- 07_Discrete_Continuous_Bridge.md
- supplements/07_INTERNAL_REASONING_SUPPLEMENT.md

Cross-corpus evidence classes:

- CAM_CLAIM_100_SCORE_AUDIT.md
- NSIT_TOOL_CLOSURE_MATRIX.md
- NSIT_REASONED_CLOSURE_PASS.md
- PAPER_REWORKS_CRYSTAL_PROJECTION.json
- standards, glue, window, and node databases where applicable
- notebook-derived review prompts and orbital source manifests

3. Definitions

- **Sample-preserving bridge.** A continuous presentation that matches every declared source sample exactly.
- **Retraction.** A map that returns an admitted representative and is stable under repeated admission.
- **Error budget.** A declared norm, tolerance, quantizer, and stability condition for non-exact bridge claims.
- **False continuity.** A continuous-looking presentation that lacks a dynamics law or sampling model.

4. Formal Claims

Claim	Lane	Statement
Theorem 8.1	standard_theorem_citation_bound_result	Every finite numeric trace admits a piecewise-linear interpolant preserving declared samples.
Theorem 8.2	normal_form_result	A valid discrete-continuous bridge preserves sample identity and provenance but not between-sample dynamics by default.
Protocol 8.3	external_calibration_result	Physical or numerical dynamics claims require an equation, sampling model, error norm, or measured trace receipt.

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 8.1	formal construction	Every finite numeric trace admits a piecewise-linear interpolant preserving declared samples	named external theorem, definition layer, citation target, or imported standard formalism	imports the external object as-is; does not silently reprove or redefine it	attach final citation metadata and mapping rule
Theorem 8.2	formal construction	A valid discrete-continuous bridge preserves sample identity and provenance but not between-sample dynamics by default	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 8.3	formal construction	Physical or numerical dynamics claims require an equation, sampling model, error norm, or measured trace receipt	dataset, measured value, uncertainty, comparator, calibration protocol, or external benchmark	only the calibrated target, scale, units, and comparator stated in the row	attach dataset/citation, uncertainty, pass/fail criterion, and falsifier

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-08 defines or uses Discrete-Continuous Bridge Without Physical Overclaim as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

5. Paper-Specific Development

1. Begin with a finite sequence of sampled values and their receipt identity.

- 2. Construct the piecewise-linear path through adjacent samples.
- 3. Declare which data are preserved exactly: source index, sample value, and segment endpoint.
- 4. Require additional dynamics claims to state an equation, sampling model, error norm, or calibration receipt.

Proof Sketch

Piecewise-linear interpolation is standard: consecutive samples define line segments that meet at shared endpoints. This guarantees sample preservation and endpoint continuity. It says nothing about the true between-sample dynamics unless a model or measurement law is supplied.

6. Construction

The construction is intentionally staged. First, the paper names the finite or typed object that can be inspected directly. Second, it declares the admissible operations over that object. Third, it records the receipt or source class that allows another paper to consume the result. Fourth, it names the residue that must not be silently promoted.

For this paper, the operative object is Discrete-Continuous Bridge Without Physical Overclaim. Its safe consumption rule is:

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source object -> declared operation -> receipt/lane -> downstream import
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The unsafe consumption rule is:

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source phrase -> downstream identity claim
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That second pattern is explicitly rejected by FLCR-01 and must be rewritten as a claim-lane promotion with evidence.

Recent External Quantum Evidence Intake

These rows add newly routed external physics papers as citation-bound or calibration-bound evidence. They are not CAM receipts unless a later tool run reconstructs their signatures.

Source	Lane	How It Helps This Paper	Boundary
EXT-RQM-CONSISTENT-2026: Quantum mechanics based on real numbers: A consistent description; DOI 10.1103/4k13-sdjh; arXiv 2503.17307	standard_theorem_citation_bound_result	Strengthens bridge-language around label changes, quotient/equivalence structure, and preserving predictions under a changed formal description.	The imported lesson is formal equivalence under stated postulates, not a generic permission to erase phase, locality, or tensor-product constraints.

7. Evidence And Receipts

Evidence source	What it contributes
Paper 07 legacy body	Sample-preserving bridge, MDHG/SpeedLight retraction, O3 structural universality.
Internal supplement 07	Interpolation/dynamics boundary, retraction, quantization error budget, aliasing guard.
CAM Claim 100 Score Audit	Paper 07 scored 90 internally closed after runtime projection/retraction support.
NSIT reasoned families	Symbolic-vs-physical carrier split and electron-hole-exciton accounting routes.

Appendix C. Implementation Intake

This addendum binds implementation work that was not fully represented in the earlier paper routing. The rows below are not pasted source text; they are evidence-lane imports that change what the paper can claim now.

Implementation source	Lane	Claim effect	Boundary
IMPL-ANALOG-FORGE-WORKBENCH	receipt_bound_internal_result	Converts the analog kit from a doctrine into a literal Python/PDF workbench: physical kit manifest, simulation layer, workbook sheets, receipts, and reports. Verification: Targeted pytest passed: 3 analog workbench tests, including eightfold manifest count, triadic binding, and demo receipt validation.	Closes kit-generation and replay scaffolding; individual paper exercises still need paper-local examples.

The immediate paper effect is stronger claim posture where these implementation rows satisfy the relevant lane. Remaining caution is reserved for specific claims that still lack an adapter, comparator, calibration target, or rerun receipt.

Appendix D. Resolved-State Closure Pass

This pass removes false restrictions on the paper's claim posture. A row is no longer called open merely because a stronger future claim exists. The satisfied lane is closed now; only the stronger claim that lacks its required adapter, comparator, calibration, or falsifier remains as residue.

Closed Now

Row	Lane	Resolved state	Remaining boundary
paper lift enumeration	receipt_bound_internal_result	8/8 LCR states succeeded; error walls=0; avg TarPit mass=0.507901.	This closes the paper-local finite lift readout, not every stronger physical interpretation.
bridge/runtime intake	normal_form_result	The bridge action is closed as a correspondence and intake surface with explicit comparator boundaries.	The family closure is a structural lane; measured physics still requires destination-specific calibration.
decade-1 tower	receipt_bound_internal_result	The decade tower is resolved with avg TarPit mass=0.510370 and mass sum=40.829567.	The decade total is an internal tower metric, not a standalone universal physical constant.
family-08 cross-lift comparison	cam_crystal_reapplication_result	Family tower binds FLCR-08, FLCR-18, FLCR-28, FLCR-38; avg TarPit mass=0.510781; error walls=0.	Cross-lift agreement strengthens routing and comparison; it does not erase paper-local boundaries.
discrete-continuous bridge	normal_form_result	The bridge is closed as an admissible correspondence protocol.	Literal continuum physics still requires limit/calibration rows.

Claim Posture After This Pass

FLCR-08 should now be read as a resolved-state contribution for Discrete-Continuous Bridge Without Physical Overclaim at its declared lane. The paper may state the strongest claim supported by these rows directly. It should reserve caution only for a specifically named stronger claim whose required evidence is absent.

8. Dependencies And Downstream Use

This paper may be imported by later FLCR papers only through the claim lanes above. The default downstream consumers are:

- FLCR-11 through FLCR-18 for window, receipt, admission, CA, algebraic, curvature, residue, and exceptional machinery.

- FLCR-28 through FLCR-30 for CAM/crystal, corpus ribbon, and universal normal-form intake.

- FLCR-31 through FLCR-40 only after the LCR-native result has been cited and the translation claim has its own evidence lane.

9. Limitations And Falsifiers

- Interpolation does not prove force laws, physical time evolution, or continuum limits.
- A continuous drawing must preserve receipt identity or it cannot be used as evidence.
- Universal encoder claims require explicit sequence-class scope.

A falsifier for this paper is any example that satisfies the declared input conditions while violating the stated construction, receipt, or lane boundary. An interpretation that merely wants a stronger downstream conclusion is not a falsifier; it is an obligation routed to a later paper.

10. Publication Readiness Tasks

- Validator identities and receipt hashes are recorded in the manifest or receipt appendix.

- External theorems require final citation entries before journal submission.
- Every theorem, proposition, and protocol must retain a claim lane and evidence boundary.
- The Markdown, LaTeX, and PDF forms are generated from the same canonical source.

Dual Positioning: Story And Formal Carrier

This paper must be read in two synchronized ways. Sequentially, it explains why the next state exists in the corpus story. Formally, it binds that state to accepted mathematics, IRL formalism, code receipts, validators, CAM/crystal lookups, and claim-lane boundaries.

Assigned original ribbon role(s): 07/bridge_action.

Original slot	Family	Lift	Role	Proof form
07	bridge_action	0	order-1 slot-7: project between discrete, continuous, lattice, or physical-facing forms	bridge, projection, calibration, or sample-preservation proof

The formal obligation is to state the strongest valid claim available for this paper without letting interpretation silently change the addressed object. Any author interpretation belongs beside the formalism as a declared relabeling, bridge, or analog, and must be accompanied by the evidence lane that makes the claim consumable by later papers.

State-Bound Proof Contract

This paper receives: state emitted by prior slot 06 and reopened at original slot 07 (Discrete Continuous Bridge).

It must produce: formal result of original slot 07 plus the same-family lift path toward slot 17.

Semantic transition: project the state between discrete, continuous, lattice, or physical-facing forms.

Accepted formalism targets to bind in the journal rewrite:

- projection maps
- discretization and sampling theory
- interpolation/continuum limits
- calibration boundary statements

Slot	Family	Lift	Produced proof form
07	bridge_action	0	bridge, projection, calibration, or sample-preservation proof

The formal carrier uses these accepted tools while preserving interpretive claims as explicit relabelings, correspondences, or bridges. Claim promotion is admissible only when a receipt, standard citation, CAM/crystal reapplication, normal-form result, calibration datum, or falsifier boundary is attached.

Provenance And Crystal Evidence Integration

This section integrates prior work, CAM/crystal projections, NSIT tooling, standards-window evidence, and routed supplements as provenance-bearing evidence. Source material is not treated as manuscript text; it is bound into the paper only through claim lanes, receipts, validators, normal forms, calibration rows, or explicit falsifier boundaries.

Expansion Rule For This Slot

Use past work to build correspondence tables: discrete source, continuous/physical target, standard theorem, calibration unit, and non-promotion guard.

Primary Evidence Inputs

Source	Placement reason
07_DiscreteContinuous_Bridge.md	primary assigned source for the paper's formal spine

Source	Placement reason
supplements/07_INTERNAL_REASONING_SUPPLEMENT.md	paper-local reasoning supplement contributing to definitions, proof sketch, and limitations

Secondary And Orbital Evidence Inputs

Source	Placement reason
supplements/RECEIPT_VERIFIER_CATALOG.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_PYRAMID_INDEX.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_5P_WINDOW_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_7P_WINDOW_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_9P_WINDOW_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
CAM_CLAIM_100_SCORE_AUDIT.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
NSIT_TOOL_CLOSURE_MATRIX.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
NSIT_REASONED_CLOSURE_PASS.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
ORBITAL_DATA_CROSS_POPULATION_AUDIT.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
AGENT_CRYSTAL_WORK_HARVEST.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
PAPER_REWORKS_CRYSTAL_PROJECTION.json	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
NEW_PAPER_NEEDS.md	route relevant obligations into this paper's burden table: NP-04 F4 Encoder Universality And Exceptional Route Conditions
Additional source routes	Complete routing is maintained in the source-placement appendix

Crystal And Standards Evidence To Bind

- Paper Reworks crystal projection: 33 paper markdown files, 9 supplements, 5 queues, 6 lattice-forge unification artifacts, 140 faces, 140 vignettes, and 280 CAM nodes.
- Standards-window intake: 192/192 corpus conformance verdicts PASS; 140/142 obligations have candidate routes; 2/4/8/16/32 window lattice is available for cross-paper reads.
- Agent/crystal harvest: agent-generated memory, runtime standards starter, current runtime build, repo-harvest CAM, notebook-derived bundles, and downloaded crystal payloads are orbital evidence surfaces.
- NSIT inventory baseline: 220 validators, 1786 receipt/data artifacts, 1137 formal-paper-like artifacts, 114 supplements, and 86 memory/CAM artifacts.

Paper-Specific CAM/Score Rows

07	90	internally closed	sample-preserving bridge, MDHG/SpeedLight retraction, O3 structural closure	between-sample physics and general F4 encoder theorem
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Paper-Specific NSIT Closure Rows

No direct NSIT row matched the assigned legacy papers in the first-pass matrix; validator matching by object name remains a receipt-attachment requirement.

Source Integration Summary

The legacy source and internal reasoning supplement are treated as source evidence rather than manuscript prose. Their contributions enter this paper as claim-lane entries, receipt or validator references, finite-domain statements, and limitation

boundaries. Speculative or cross-domain interpretations remain separate from closed local results until an explicit adapter, citation, normal-form derivation, calibration row, or falsifier is attached.

Claim Binding Queue

This queue records the evidence discipline for claim strengthening. It separates claims that already have support from residues that remain in falsifier or open-obligation lanes.

Queue	Lane	Promote now	Bind next	Residue
Symbolic Carrier Versus Physical Carrier	external_calibration_result	Promote addressable symbolic-carrier and transport claims where the paper names carrier, path, residue, or traversal rules.	Map every physical carrier phrase to a standard physics object, Hamiltonian/model, dataset, or calibration route before treating it as measured physics.	Physical identity, units, material constants, and measured transport remain calibration-bound.
Electron-Hole-Exciton Standard Accounting	standard_theorem_citation_bound_result	Treat hole, vacancy, exciton, recombination, screening, band-gap, and effective-mass language as standard-model correspondence when a material model is present.	Attach standard equations/citations and state what LCR adds: addressability, obligation routing, and residue bookkeeping.	Actual material behavior and quantitative exciton claims need a material model, units, and data.
Continuum Edge And Sampling-Stability Bridge	external_calibration_result	Promote discrete-continuous bridge claims only when the sample rule, norm, error bound, or counterexample policy is stated.	Attach sampling rule, interpolation/limit statement, norm, error bound, units, and boundary condition.	Loose continuum, GR, plasma, or energy language stays presentation-level until calibrated.

Binding Discipline

Each queue item is represented as a delta:

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local claim at paper time
-> attached later source/tool/receipt/theorem
-> revised representation
-> invariant boundary and remaining residue
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This preserves the sequential story while allowing later tools, crystals, and standard formalisms to return through the proper lane.

Claim Delta Ledger

This ledger applies the paper's claim-binding queues as explicit back-application deltas. It keeps the sequential paper story intact while allowing later receipts, standard formalisms, CAM/crystal routes, and validators to sharpen the claim.

Delta	Local claim at paper time	Later evidence	Revised representation	Boundary kept
Symbolic Vs Physical Carrier	This paper may use carrier language for addressable symbolic transport inside the LCR/CAM system.	Standard physics carrier terminology, local verifier receipts, material/plasma/transport supplements, and calibration routes.	Split symbolic carrier closure from measured physical carrier identity.	Physical identity, units, Hamiltonians, datasets, and measured constants remain external-calibration claims.
Electron Hole Exciton Accounting	Vacancy, complement, hole, residue, and excitation language may appear as LCR addressability/residue vocabulary.	Standard electron-hole-exciton formalism, material-model rows, band-gap/screening/recombination equations, and NP-12 routing.	Treat hole/exciton vocabulary as standard correspondence where a material model exists, while preserving LCR's contribution as addressability and obligation bookkeeping.	Quantitative material behavior requires model parameters, units, citations, and data.
Continuum Sampling Bridge	This paper may bridge discrete and continuous language only as far as its sampling, projection, or boundary rule supports.	Sampling/interpolation rules, norms, error bounds, boundary conditions, units, and calibration routes.	Promote bridge claims only when sample rule, norm, and error or counterexample policy are stated.	Loose continuum, GR, plasma, or energy language remains presentation-level or calibration-bound.

The revised representation records the strongest claim supported by the current evidence package. Promotion into theorem or proposition language occurs only when the named evidence is attached in the manifest or workbook replay.

Source-Backed Evidence Summary

Routed archive and mirror cards are treated as provenance summaries rather than body text. They identify candidate receipts, validators, theorem registries, or supplemental source routes. A card strengthens this paper only when its contribution is bound to a claim lane, evidence object, and boundary.

Evidence card	Score	Publication contribution	Boundary
CQE-paper-08.50: Paper 8.50 - Lattice Closure Claim Contract	81	Paper 8.50 lets later papers import the lattice closure scaffold honestly. It keeps the verified local rungs useful while preserving global landings as separate proof obligations.	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
FINAL_FORMAL_PAPER: Complete Formal Claims: The Folded Strand	75	We present the complete closed-form claim set of the CQE_CMLPX corpus: - 33 papers = 33 folding operations on a self-complementary strand - 144 tools = cumulative analog kit = digital twin surface - 135 digital twins ...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
SUMMARY-V-32-Theorems-Registry: Summary Paper V — The 32 Theorems in One Table: Closed-Form Registry	75	This paper is the complete closed-form registry of all 32 theorems in the CQE_CMLPX corpus. Each theorem is stated precisely, given its formal context (where it is proven), and listed with its verifier. The table ...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
CQE-FORMAL-07_META_CORPUS: CQECMLPX FORMALIZATION PAPER 7	69	The meta-packaging layer IS the LCR triality recognizing itself. The Grand Ribbon frames the corpus, the Meta-LCR reads it, the Supervisor Cursor schedules it. All three are the same triality: the corpus as C, its...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
block-01-dyad-02-07: Block 01 Dyad Brief - Papers 2 and 7	67	Source: Dyad Agent B, read-only synthesis. Agent inspected files and ran verifiers. No edits made by agent. Final scientific focus: Paper 2 is the first rigorous residue theorem. The integer paper should center on the...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
Additional evidence cards	14 total	The complete card inventory is maintained in the archive evidence matrix.	Cards are source-discovery aids until bound to paper-local evidence.

Journal Apparatus

Article Type And Intended Venue Fit

This manuscript is written as a formal-methods and mathematical-systems paper inside the series *Formalizing LCR, Unifying Standard Models*. Its intended reader is a technical reviewer who needs claim boundaries, reproducibility routes, and cross-paper dependencies to be visible without relying on private context.

Structured Contribution Statement

Item	Statement
Publication ID	FLCR - 08
Legacy source slot(s)	07
Ribbon role	order-1 slot-7: project between discrete, continuous, lattice, or physical-facing forms
Proof form	bridge, projection, calibration, or sample-preservation proof
Received state	state emitted by prior slot 06 and reopened at original slot 07 (Discrete Continuous Bridge)
Produced state	formal result of original slot 07 plus the same-family lift path toward slot 17

Claim-Evidence Table

Claim	Lane	Evidence to attach	Boundary
Theorem 8.1	standard_theorem_citation_bound_result	Receipt, source card, validator, citation, or CAM route named in the paper manifest	Every finite numeric trace admits a piecewise-linear interpolant preserving declared samples.

Claim	Lane	Evidence to attach	Boundary
Theorem 8.2	normal_form_result	Receipt, source card, validator, citation, or CAM route named in the paper manifest	A valid discrete-continuous bridge preserves sample identity and provenance but not between-sample dynamics by default.
Protocol 8.3	external_calibration_result	Receipt, source card, validator, citation, or CAM route named in the paper manifest	Physical or numerical dynamics claims require an equation, sampling model, error norm, or measured trace receipt.

Figure Plan

Figure	Purpose	Caption
FLCR-08-F1	Correspondence bridge	Schematic showing how FLCR-08 turns its received state into the produced state while preserving claim lanes and residue boundaries.
FLCR-08-F2	Evidence routing map	Diagram of source papers, archive cards, CAM/crystal routes, validators, and workbook replay paths used by this manuscript.
FLCR-08-F3	Same-family lift context	Diagram placing this paper in the original 00 - 79 ribbon family and showing predecessor/successor slots.

In-Text Figure FLCR-08-F1: Correspondence bridge

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received state
-> Correspondence bridge
-> formal claim lane
-> evidence object / receipt / citation
-> produced state
-> remaining residue

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Table Plan

Table	Purpose
FLCR-08-T1	Source-target correspondence table
FLCR-08-T2	Claim-lane/evidence-boundary table
FLCR-08-T3	Archive evidence card and source-backed upgrade table
FLCR-08-T4	Workbook replay and falsifier table

Worked Example

Translate one discrete source claim into a continuous/standard-model-facing target with boundary. The example must name the input object, the operation, the evidence object, the allowed revised claim, and the remaining boundary.

Nomenclature And Glossary

Term	Paper-local meaning
CAM	Defined by this paper as part of the bridge_action proof role; final definition is recorded in the paper-local glossary.
bridge	Defined by this paper as part of the bridge_action proof role; final definition is recorded in the paper-local glossary.
calibration	Defined by this paper as part of the bridge_action proof role; final definition is recorded in the paper-local glossary.
claim lane	Defined by this paper as part of the bridge_action proof role; final definition is recorded in the paper-local glossary.
correspondence	Defined by this paper as part of the bridge_action proof role; final definition is recorded in the paper-local glossary.
crystal	Defined by this paper as part of the bridge_action proof role; final definition is recorded in the paper-local glossary.
projection	Defined by this paper as part of the bridge_action proof role; final definition is recorded in the paper-local glossary.
receipt	Defined by this paper as part of the bridge_action proof role; final definition is recorded in the paper-local glossary.

Appendix FLCR-08-A: Evidence Package

The appendix records:

- 1. Legacy source excerpts or source-card references used in the final claim ledger.
- 2. Receipt and verifier paths, including pass/fail/partial status.
- 3. CAM/crystal query or projection rows used by the paper, with source identity and boundary.
- 4. Standards-window and NSIT evidence used for conformance or routing.
- 5. Falsifier, calibration, or external-data rows that remain unresolved.

Data And Code Availability

The publication package contains the canonical Markdown sources, manifests, source-routing matrices, validation reports, and generated PDF/Tex outputs.

Code and verifier references are cited through manifests, archive evidence cards, receipt catalogs, or workbook replay tables. External datasets or standards citations must be attached before any calibration-bound claim is promoted.

Reviewer Checklist

Check	Reviewer question
Claim lane	Does every strong claim declare its lane and evidence type?
Boundary	Does the paper preserve what it does not prove?
Reproducibility	Is there a receipt, verifier, source card, citation, or replay table for the claim?
Cross-paper import	Does the paper cite the prior FLCR/OPR slot before consuming its result?
Interpretation	Does the analog layer preserve the addressed state while changing labels/access paths?

Declarations

No human-subjects, clinical, or animal-research protocol is asserted by this paper. External empirical claims require separate dataset, calibration, or domain-validation attachments. Competing-interest and funding statements are recorded in the submission metadata.

11. Publication Integration Addendum

11.1 Role In The Series

FLCR-08 (Discrete-Continuous Bridge Without Physical Overclaim) occupies the bridge_action role at lift depth 0. The paper receives state emitted by prior slot 06 and reopened at original slot 07 (Discrete Continuous Bridge). It produces formal result of original slot 07 plus the same-family lift path toward slot 17. The state transition is: project the state between discrete, continuous, lattice, or physical-facing forms.

The paper-local proof form is:

bridge, projection, calibration, or sample-preservation proof

The integration result is a correspondence map with explicit calibration and falsifier boundaries. This addendum records how that result is consumed by the publication series without relying on editorial instructions or private working context.

11.2 Formal Carrier

Formal carrier	Function
projection maps	Formal carrier for the paper-local result.
discretization and sampling theory	Formal carrier for the paper-local result.
interpolation/continuum limits	Formal carrier for the paper-local result.
calibration boundary statements	Formal carrier for the paper-local result.

The LCR, CAM, crystal, forge, and analog vocabulary functions as a typed access layer over these carriers. A relabeling is admissible only when the addressed object, evidence lane, boundary, and downstream import rule remain attached.

11.3 Claim Ledger

Claim	Lane	Statement
Theorem 8.1	standard_theorem_citation_bound_result	Every finite numeric trace admits a piecewise-linear interpolant preserving declared samples.
Theorem 8.2	normal_form_result	A valid discrete-continuous bridge preserves sample identity and provenance but not between-sample dynamics by default.
Protocol 8.3	external_calibration_result	Physical or numerical dynamics claims require an equation, sampling model, error norm, or measured trace receipt.

These claims are consumed at their stated lane. A stronger downstream statement creates a new claim envelope with its own evidence object.

11.4 Evidence Package

Evidence class	Routed items	Status
Legacy sources	07_Discrete_Continuous_Bridge.md, supplements/07_INTERNAL_REASONING_SUPPLEMENT.md	routed evidence
Evidence inputs	CAM_CLAIM_100_SCORE_AUDIT.md, NSIT_TOOL_CLOSURE_MATRIX.md, NSIT_REASONED_CLOSURE_PASS.md, PAPER_REWORKS_CRYSTAL_PROJECTION.json	routed evidence
CAM/crystal sources	PAPER_REWORKS_CRYSTAL_PROJECTION.json, AGENT_CRYSTAL_WORK_HARVEST.json, runtime standards artifact, notebook-derived source bundle	routed evidence
Standards-window inputs	window_summary.json, node DBs, window DBs	routed evidence
Receipts	external requirement	not attached in this source package
Validators	external requirement	not attached in this source package

Standards-window, runtime, and notebook-derived artifacts are treated as evidence-routing surfaces. They support source discovery, conformance checking, and reapplication, but they do not replace citations, receipts, normal-form derivations, calibration rows, or falsifiers.

11.5 Results Representation

The paper's results section is organized around five publication objects:

1. the exact object acted on at this slot;
2. the admissible operation or transformation;
3. the receipt, enumeration, derivation, lookup, or external theorem supporting the operation;
4. the residue or boundary left after the result;
5. the downstream import rule.

11.6 Figures And Tables

Figure	Role
FLCR-08-F1: Correspondence bridge	Visualizes the proof object, routing, or boundary.
FLCR-08-F2: Evidence routing map	Visualizes the proof object, routing, or boundary.
FLCR-08-F3: Same-family lift context	Visualizes the proof object, routing, or boundary.

Table	Role
FLCR-08-T1: Source-target correspondence table	Records claim lanes, evidence, sources, or falsifiers.

Table	Role
FLCR-08-T2: Claim-lane/evidence-boundary table	Records claim lanes, evidence, sources, or falsifiers.
FLCR-08-T3: Archive evidence card and source-backed upgrade table	Records claim lanes, evidence, sources, or falsifiers.
FLCR-08-T4: Workbook replay and falsifier table	Records claim lanes, evidence, sources, or falsifiers.

11.7 Limits And Falsifiers

The paper proves only the object and domain declared in its claim ledger. Physical, Standard Model, observer, calibration, or synthesis claims require the additional lane evidence stated by their destination papers. CAM/crystal and forge language is operational only when represented as an address, lookup, receipt, validator, or reapplication path.